

Yuxuan Wang

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SUMMARY

Computational and Applied Physics Ph.D. candidate and French-certified Engineer (Titre d'Ingénieur), specializing in data analysis, machine learning, and large-scale scientific signal processing. Experienced in Python-based analysis pipelines and hydrodynamic simulations for extracting weak or transient signals from noisy astrophysical datasets. Skilled in building automated workflows for data-intensive research and developing physical models for complex systems.

EDUCATION

Universitat Politècnica de Catalunya (UPC) Ph.D. in Computational and Applied Physics (GAA)	Spain Oct. 2024 – Present
• Focus: Numerical simulation of hydrodynamic systems and high-energy transients.	
Ecole Centrale de Lille Diplôme d'Ingénieur (Master of Engineering - CTI Accredited)	France Sept. 2019 – Sept. 2021
• Selective engineering program emphasizing advanced mathematics, programming and system engineering.	
Xi'an Jiao Tong University (C9 / Project 985) M.A. in Electronic & Information Engineering B.A. in Biomedical Engineering	China Sept. 2021 – Sept. 2024 Sept. 2017 – Sept. 2021

RESEARCH & TECHNICAL EXPERIENCE

Lead Researcher Numerical Simulation of Rare Hydrodynamic Transients <i>Ph.D. Project</i>	UPC, Spain Oct. 2024 – Present
• Executed high-fidelity implicit hydrodynamic simulations computing the SHIVA code to model extreme thermonuclear runaways (Helium Novae), computing solutions for distinguishing these events from their hydrogen-burning analogues.	
• Architected end-to-end automated data pipelines using Python to parse massive HPC simulation outputs, enabling rapid parameter-space exploration and time-series extraction.	
• Aimed at establishing future observational diagnostics for helium novae.	
• Scientific Communication: Presented research via talks and posters at international conferences (e.g., XVIII International Symposium on NUCLEI IN THE COSMOS), collaborating with global researchers to validate theoretical models against observational data.	
Algorithm Researcher Data & Signal Processing <i>National Time Service Center, Chinese Academy of Sciences</i>	CAS, China Mar. 2023 – Jun. 2024
• Implemented deep learning models to optimize Fast Radio Burst (FRB) searching, accelerating the sifting of rare signals from petabytes of noisy observation data.	
• Based on the 40-meter radio telescope , deployed Linux-based pipelines (PRESTO, HEIMDALL, TransientX) to search for single pulses and FRBs, and to think about methods to obtain high-quality observation data.	
• Engineered algorithms to mitigate Radio Frequency Interference (RFI), improving the Signal-to-Noise Ratio (SNR).	
AI & Game Developer Monte Carlo Optimization & Websites <i>AWALE, Ecole Centrale de Lille (awale.rezoleo.fr)</i>	Lille, France Nov. 2019 – Feb. 2021
• Applied Monte Carlo algorithms to design and predict states for artificial intelligence games.	
• Integrated front-end and back-end systems, ensuring robust performance and real-time state calculation.	
Medical Device Assembly & Administration Intern <i>Lille, France T.L.V. Healthcare</i>	Jan. 2020 – Feb. 2020

SKILLS & HONORS

- **Programming:** Python (PyTorch, TensorFlow, Pandas, NumPy), Fortran, Matlab, Linux/HPC, HTML/CSS.
- **Languages:** English (TOEIC: 920/990), French (Delf B2), Mandarin (Native).
- **Awards:** CSC Scholarship ×6 (2019-21, 2024-28); Xi'an Jiaotong University Scholarship ×4.

LEADERSHIP & EXTRACURRICULAR

- **Art Director & Deputy Art Troupe Leader:** Managed university-level events and coordinated group decisions at XJTU Student Art Troupe (2021-24).
- **Secretary:** Chinese Culture Club (France, 2020-21); promoted international exchange, integrated diverse student groups and organized activities.